

HINTS AND SOLUTIONS

- Q.1 A
Q.2 B
Q.3 D
Q.4 D
Q.5 C
Q.6 A
Q.7 A
Q.8 B
Q.9 C
Q.10 D

Sol. Once the body acquire the terminal velocity it will have uniform motion.

$$v_t = \frac{2}{9} \frac{r^2}{\eta} [(\rho - \sigma)g], \text{ down}$$

$$v_t = \frac{40}{9} \text{ m/s down}$$

$$\text{Distance travelled} = \frac{40}{9} \times 0.1 = \frac{4}{9} \text{ m}$$

- Q.11 C

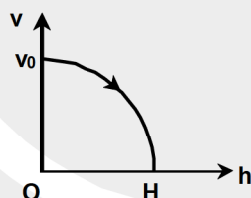
Sol. Reason: Streamlines may be straight or curved.

- Q.12 A

Sol. Applying Bernoulli's theorem

$$\frac{1}{2} \rho v^2 = \rho g(H - h)$$

$$v^2 = 2g(H - h)$$



- Q.13 C

$$\text{Sol. } P_0 + 0.8915 + \frac{1}{2} \rho v^2 + 0 = (P_0 + 0.895) + 0 + 1910$$

- Q.14 A

$$\text{Sol. } P_0 + 0.8915 = P_0 + 0.895 + 1 \times g \times 4$$

- Q.15 B

$$\text{Sol. } \sqrt{204 - 160} \text{ a} = A(-dy/dt)$$

Q.16 [Ans. (A) P, S, (B) P, Q, (C) R, (D) P,Q,S]

Sol. If A and C are open air come from A due to atm per.

(A) and liquid will come out due to per difference

(B) B and C are open air will come from lower per

(C) If only C is open small amount of water will come out hil per on both sides is same

Q.17 Ans $\sqrt{\frac{6T}{\rho} \left(\frac{1}{r} - \frac{1}{R} \right)}$

Q.18 Ans $\frac{\omega^2 L^2}{2g}$

Q.19 Ans. $\frac{4}{3} \pi r^3 \rho_w g \left[\frac{2r^4 g}{81 \eta^2} (\rho_w - \rho_{air})^2 - h \left(1 - \frac{\rho_{air}}{\rho_w} \right) \right]$

Sol. $W_{air} + W_{Buoyancy} + W_g = \Delta KE$

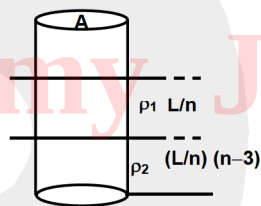
For terminal velocity use, $mg = F_{Buoyancy} + 6 \pi \eta r v$

Q.20 Ans. $\left[\rho_2 + \frac{\rho_1 - 3\rho_2}{n} \right]$

Sol. Apply the following equation

$$\rho A L g = \rho_1 A \frac{L}{n} g + \rho_2 A g \frac{L}{n} (n-3)$$

$$\therefore \rho = \left[\rho_2 + \frac{\rho_1 - 3\rho_2}{n} \right]$$



Q.21 Ans. $P = \frac{\pi \eta \omega^2 R^4}{2t}$

Sol. Let coefficient of viscosity & thickness of film $\rightarrow \eta$ & x respectively

Viscous force on element ring is

$$dF = \eta (2\pi r dr) \left[\frac{wr - 0}{x} \right]$$

$$d\tau = (dF) r$$

$$\therefore \tau = \frac{\eta 2\pi w}{t} \int_0^R r^3 \cdot dr = \left(\frac{\eta (2\pi) w}{4x} \right) R^4$$

$$\therefore \tau \propto R^4]$$

